

Full Stack Development with MERN

Final Report format

BookNest – Where Stories Nestle

(MERN Stack Online Book Store)

Introduction

◇ Project Title: BookNest – Where Stories Nestle

◇ Team Members:

Name	Role
Team Leader	Full Stack Developer (Frontend + Backend)
Team Member 2	UI/UX Designer
Team Member 3	Backend Developer
Team Member 4	Database & Deployment Engineer

Project Overview

◇ Purpose:

BookNest is a full-stack MERN application designed to provide a secure, user-friendly online platform for browsing, searching, and purchasing books. The goal is to create a modern, scalable bookstore with smooth navigation and secure authentication.

◇ Key Features:

- User Registration & Login (JWT Authentication)
- Browse 100+ Books with images
- Search books by title or author
- Add to Cart functionality
- Place Orders
- View Order History
- Cloud Deployment (Render + Vercel)
- Secure MongoDB Atlas database

Architecture

◇ Frontend (React.js – Vite)

- Component-based architecture
- Pages: Home, Register, Login, Books, Cart, Orders
- Axios used for API calls
- Protected routes for authentication

- Hosted on Vercel
- ◇ Backend (Node.js + Express.js)
 - RESTful API structure
 - JWT authentication
 - Middleware for validation
 - CORS configured for production
 - Hosted on Render
- ◇ Database (MongoDB Atlas)

Collections:

- Users
- Books
- Orders

Database Interactions:

- Mongoose models
- CRUD operations
- Cloud-hosted NoSQL database

Setup Instructions

- ◇ Prerequisites
 - Node.js (v18+)
 - MongoDB Atlas account
 - Git
 - npm

◇ Installation

Step 1: Clone Repository

```
git clone https://github.com/Kushulatha-k/BookNest.git
cd booknest
```

Step 2: Backend Setup

```
cd backend
```

```
npm install
```

Create .env file:

```
PORT=5000
```

```
MONGO_URI=your_mongodb_connection_string
```

```
JWT_SECRET=your_secret_key
```

Run backend:

```
npm run dev
```

Step 3: Frontend Setup

```
cd frontend
```

```
npm install
```

```
npm run dev
```

Folder Structure

◇ Client (Frontend)

```
frontend/
```

```
|— src/
|   |— components/
|   |— pages/
|   |— api/
|   |— App.jsx
|   |— main.jsx
|— package.json
```

◇ Server (Backend)

```
backend/
```

```
|— models/
|— routes/
|— config/
|— server.js
|— package.json
|— .env
```

Running the Application

◇ Frontend

```
cd frontend
```

```
npm run dev
```

Runs at:

```
http://localhost:5173
```

◇ Backend

```
cd backend
```

```
npm run dev
```

Runs at:

http://localhost:5000

API Documentation

◆ User Routes

Method	Endpoint	Description
POST	/api/users/register	Register new user
POST	/api/users/login	Login user

Example Request:

```
{  
  "email": "user@gmail.com",  
  "password": "123456"  
}
```

◆ Book Routes

Method	Endpoint	Description
GET	/api/books	Get all books
GET	/api/books/seed	Seed books

◆ Order Routes

Method	Endpoint	Description
POST	/api/orders	Place order
GET	/api/orders/myorders	Get user orders

Authentication

- JWT-based authentication
- Token stored in localStorage
- Protected routes
- Password encrypted using bcrypt
- Middleware validates token before accessing protected APIs

User Interface

The UI includes:

- Animated backgrounds
- Responsive design
- Card-based book layout

- Navigation bar
 - Styled cart and order pages
- (Screenshots included in README)

Testing

Testing Strategy:

- API Testing using Postman
- Manual Integration Testing
- Authentication Testing
- Deployment Testing

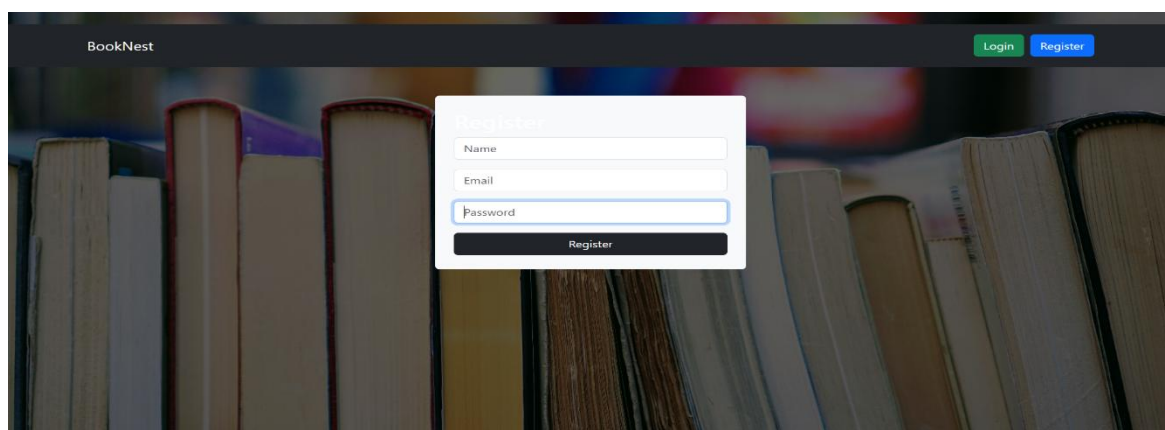
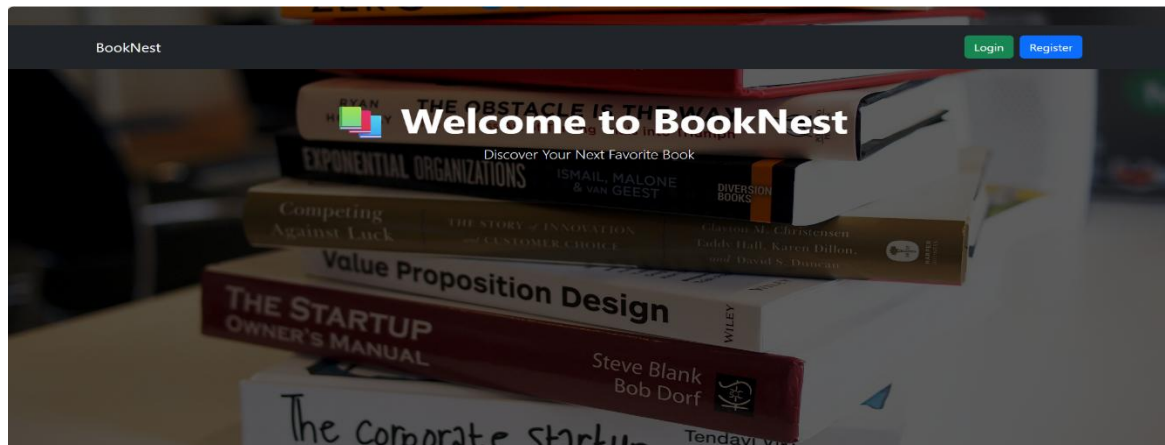
All modules tested before production deployment.

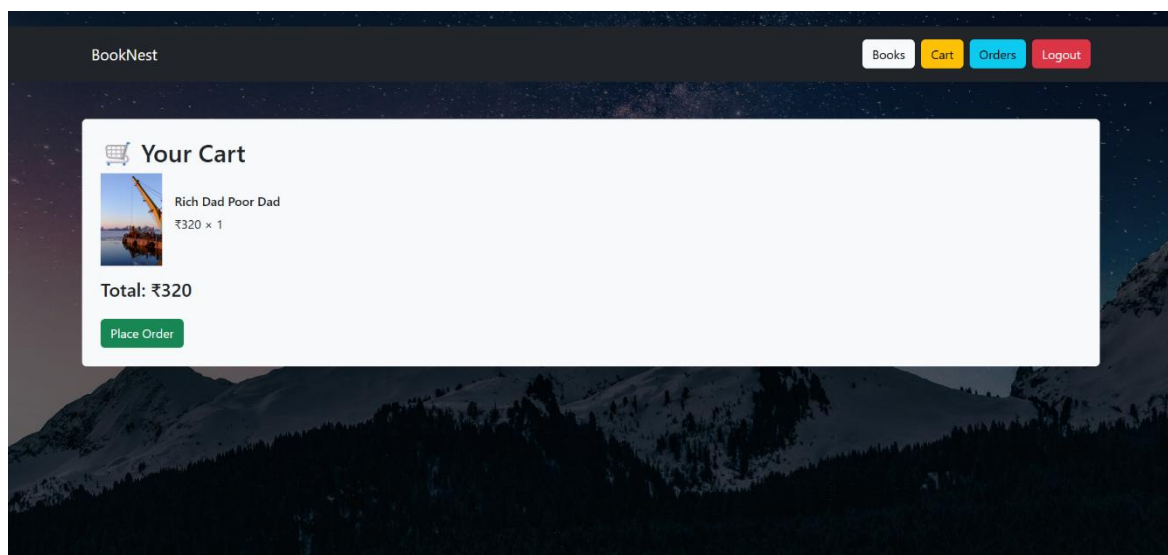
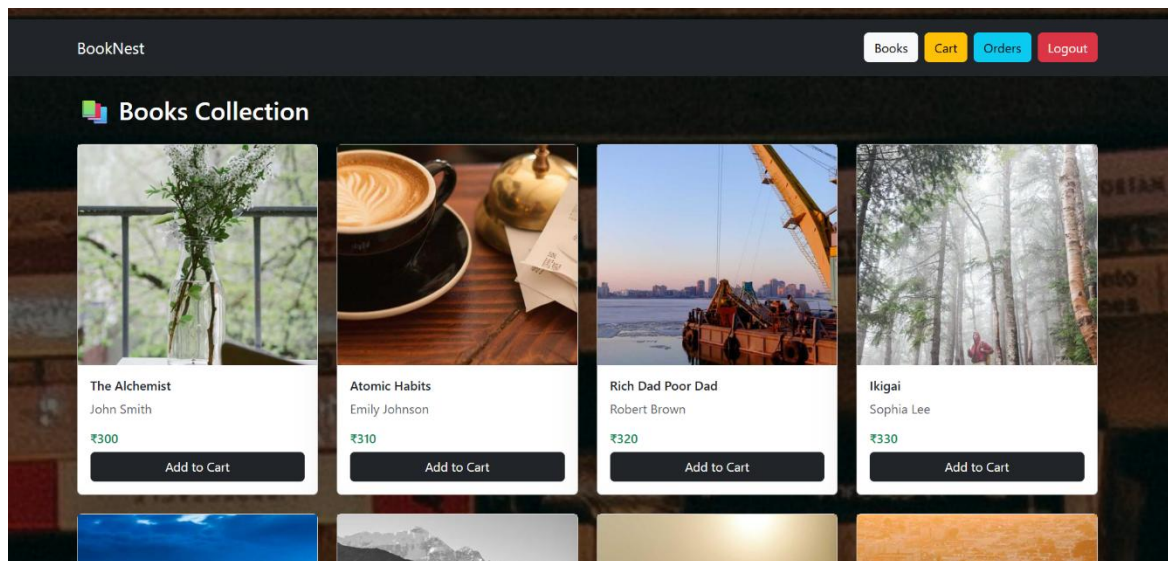
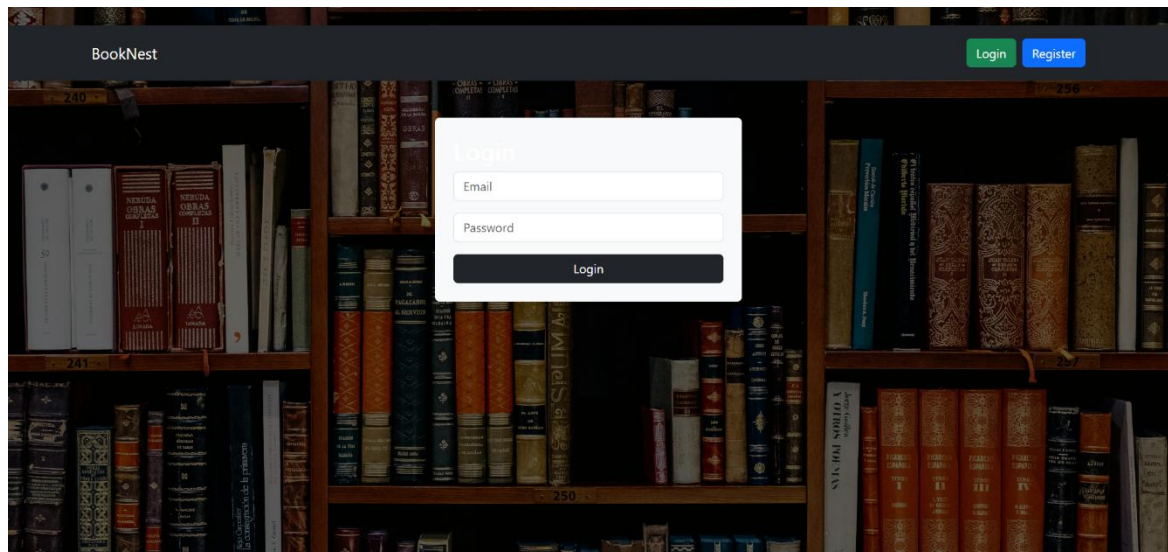
Screenshots / Demo

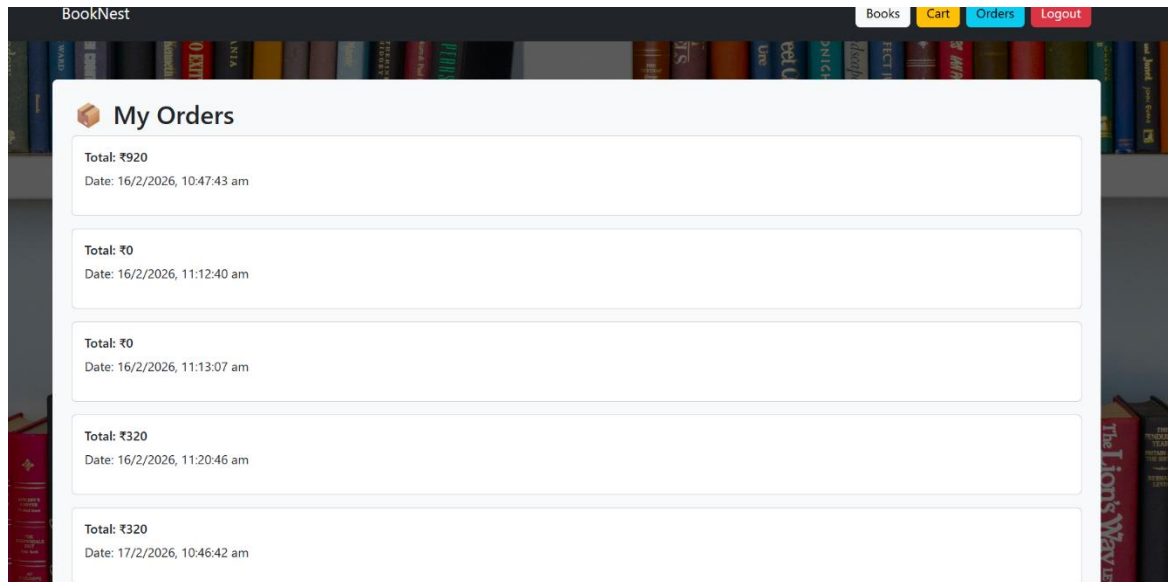
Live Demo:

Frontend: <https://book-nest-six-topaz.vercel.app/>

Backend : <https://booknest-backend-0buw.onrender.com>







Known Issues

- No payment gateway integration (currently demo order placement)
- No email verification
- Limited admin dashboard features
- Basic search (no AI recommendation yet)

Future Enhancements

- Payment Gateway Integration (Stripe/Razorpay)
- AI-based Book Recommendation System
- Admin Dashboard
- Order Tracking
- Mobile Application (React Native)
- Wishlist Feature
- Review & Rating System

Conclusion : BookNest is a scalable MERN stack application that provides a secure, modern online bookstore platform. It demonstrates full-stack development, cloud deployment, authentication, and REST API integration using modern web technologies.

APPENDIX:

Source Code(if any):

```
import Navbar from "../components/Navbar";  
import PageWrapper from "../components/PageWrapper";
```

```
function Home() {
```

```

return (
  <PageWrapper bg="https://images.unsplash.com/photo-1512820790803-83ca734da794">
    <Navbar />

    <div className="container text-center mt-5">
      <h1 className="display-4 fw-bold">📖 Welcome to BookNest</h1>
      <p className="lead">Discover Your Next Favorite Book</p>
    </div>
  </PageWrapper>
);
}

```

```
export default Home;
```

```

import { useState } from "react";
import axios from "../api/axios";
import { useNavigate } from "react-router-dom";
import Navbar from "../components/Navbar";
import PageWrapper from "../components/PageWrapper";

```

```

function Register() {
  const [form, setForm] = useState({
    name: "",
    email: "",
    password: ""
  });
}

```

```
const navigate = useNavigate();
```

```

const submitHandler = async (e) => {
  e.preventDefault();

```

```

  try {
    await axios.post("/users/register", form);
    alert("Registered Successfully ✅");
    navigate("/login");
  }

```



```

    } catch (error) {
      alert("Registration Failed ✖");
      console.log(error.response?.data);
    }
  };

  return (
    <PageWrapper bg="https://images.unsplash.com/photo-1495446815901-
a7297e633e8d">
      <Navbar />

      <div className="container mt-5 bg-light p-4 rounded" style={{ maxWidth:
"400px" }}>
        <h2>Register</h2>

        <form onSubmit={handleSubmit}>
          <input
            className="form-control mb-3"
            placeholder="Name"
            value={form.name}
            onChange={(e) =>
              setForm({ ...form, name: e.target.value })
            }
          />

          <input
            className="form-control mb-3"
            placeholder="Email"
            value={form.email}
            onChange={(e) =>
              setForm({ ...form, email: e.target.value })
            }
          />

          <input
            type="password"
            className="form-control mb-3"

```

```

        placeholder="Password"
        value={form.password}
        onChange={(e) =>
            setForm({ ...form, password: e.target.value })
        }
    />

    <button className="btn btn-dark w-100">
        Register
    </button>
</form>
</div>
</PageWrapper>
);
}

export default Register;

import { useEffect, useState } from "react";
import axios from "../api/axios";
import Navbar from "../components/Navbar";
import BookCard from "../components/BookCard";
import PageWrapper from "../components/PageWrapper";

function Books() {
    const [books, setBooks] = useState([]);

    useEffect(() => {
        const fetchBooks = async () => {
            const { data } = await axios.get("/books");
            setBooks(data);
        };
        fetchBooks();
    }, []);

    return (
        <PageWrapper bg="https://images.unsplash.com/photo-1507842217343-

```

```
583bb7270b66">
```

```
<Navbar />
```

```
<div className="container mt-4">
```

```
<h2 className="mb-4">📖 Books Collection</h2>
```

```
<div className="row">
```

```
{books.map((book) => (
```

```
<div className="col-md-3 mb-4" key={book._id}>
```

```
<BookCard book={book} />
```

```
</div>
```

```
))}
```

```
</div>
```

```
</div>
```

```
</PageWrapper>
```

```
);
```

```
}
```

```
export default Books;
```

```
import { useState, useEffect } from "react";
```

```
import Navbar from "../components/Navbar";
```

```
import axios from "../api/axios";
```

```
import { useNavigate } from "react-router-dom";
```

```
import PageWrapper from "../components/PageWrapper";
```

```
function Cart() {
```

```
  const [cart, setCart] = useState([]);
```

```
  const navigate = useNavigate();
```

```
  useEffect(() => {
```

```
    setCart(JSON.parse(localStorage.getItem("cart")) || []);
```

```
  }, []);
```

```
  const totalPrice = cart.reduce(
```

```
    (acc, item) => acc + item.price * item.quantity,
```

```

0
);

const placeOrder = async () => {
  const token = localStorage.getItem("token");

  if (!token) {
    alert("Please Login First");
    navigate("/login");
    return;
  }

  try {
    await axios.post(
      "/orders",
      {
        books: cart.map(item => ({
          book: item._id,
          quantity: item.quantity
        })),
        totalPrice
      },
      {
        headers: { Authorization: token }
      }
    );

    alert("Order Placed Successfully 🛒");

    localStorage.removeItem("cart");
    setCart([]);

    navigate("/orders");

  } catch (error) {
    alert("Order Failed ❌");
  }
}

```

```
};
```

```
return (  
  <PageWrapper bg="https://images.unsplash.com/photo-1519681393784-  
d120267933ba">  
    <Navbar />  
  
    <div className="container mt-5 text-dark bg-light p-4 rounded">  
      <h2>🛒 Your Cart</h2>  
  
      {cart.length === 0 && <p>Your cart is empty.</p>}  
  
      {cart.map(item => (  
        <div  
          key={item._id}  
          className="d-flex align-items-center mb-3"  
        >  
          <img  
            src={item.image}  
            alt={item.title}  
            width="80"  
            className="me-3"  
          />  
  
          <div>  
            <h6>{item.title}</h6>  
            <p>  
              ₹{item.price} × {item.quantity}  
            </p>  
          </div>  
        </div>  
      )  
    )  
  )  
  
  {cart.length > 0 && (  
    <div>  
      <h4>Total: ₹{totalPrice}</h4>  
    </div>  
  )  
)
```

```

        <button
          className="btn btn-success mt-3"
          onClick={placeOrder}
        >
          Place Order
        </button>
      </>
    )}
  </div>
</PageWrapper>
);
}

```

```
export default Cart;
```

```

import { useEffect, useState } from "react";
import axios from "../api/axios";
import Navbar from "../components/Navbar";
import PageWrapper from "../components/PageWrapper";

```

```

function Orders() {

  const [orders, setOrders] = useState([]);

  useEffect(() => {
    const fetchOrders = async () => {
      const token = localStorage.getItem("token");

      if (!token) return;

      try {
        const { data } = await axios.get("/orders", {
          headers: { Authorization: token }
        });

        setOrders(data);
      }
    }
  });
}

```

```

    } catch (error) {
      console.log(error);
    }
  };

  fetchOrders();
}, []);

return (
  <PageWrapper bg="https://images.unsplash.com/photo-1524578271613-
d550eacf6090">
    <Navbar />

    <div className="container mt-5 text-dark bg-light p-4 rounded">
      <h2>📦 My Orders</h2>

      {orders.length === 0 && <p>No Orders Yet.</p>}

      {orders.map(order => (
        <div
          key={order._id}
          className="card p-3 mb-3"
        >
          <h6>Total: ₹{order.totalPrice}</h6>
          <p>
            Date: {new Date(order.createdAt).toLocaleString()}
          </p>
        </div>
      ))}
    </div>
  </PageWrapper>
);
}

export default Orders;

```

GitHub : <https://github.com/vamsi746/BookNest.git>

Project Demo Link :

<https://drive.google.com/file/d/11Keyg1mScVI3XNe7mr8SzXEGlfJz96od/view?usp=sharing>